Who Counts as a Neighbor?

From geographic maps to a Python laboratory of proximity graphs

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Who decides which points are neighbors? A map of water stations, a set of image pixels, or a cloud of learned embeddings supplies points but not a unique network. Connecting every pair is usually uninformative; connecting only the nearest pairs may hide a meaningful boundary. A *proximity graph* makes that decision explicit through geometry. Our open-source **ProximityGraphs** Python library lets readers build, compare, and experiment with these competing models rather than accept a single default.

How Did Proximity Graphs Emerge?

Spatial partitions originated long before modern graph software. Voronoi formalized nearest-site regions in 1908, and Delaunay triangulations subsequently represented which regions meet. An earlier historical setting was John Snow's map of London cholera deaths in 1854: our *modern* Voronoi overlay places pump coordinates into nearest-site cells. Snow neither used Voronoi cells nor established water use by straight-line distance. Still, the example immediately poses a practical question: which facility serves which location, and when does geometric proximity misrepresent street access?

Figure 1 · John Snow's pumps: Voronoi and Delaunay

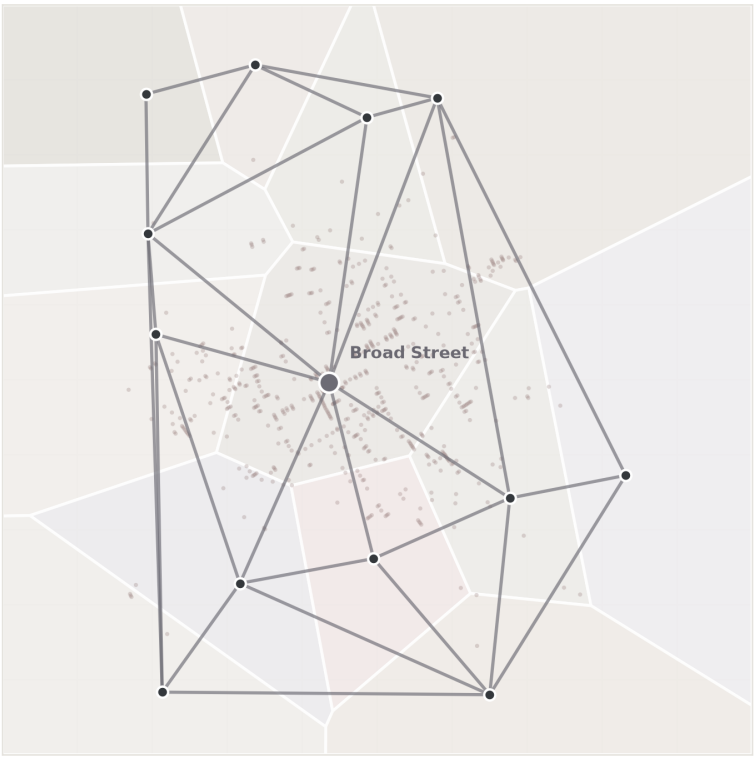


Figure 1: Historical geography and facility allocation. Thirteen digitized Soho water pumps define modern Voronoi regions and Delaunay connections; 578 cholera death locations remain contextual observations, not graph vertices. Locations: HistData digitization; artwork: the authors using ProximityGraphs. Explore the interactive figure.

In 1969, Gabriel and Sokal proposed joining sampled populations only if the diameter disk between their sites contained no third point. Toussaint (1980) replaced the disk with the intersection of two endpoint-centered disks, yielding the *relative neighborhood graph* (RNG). Alpha-shapes (Edelsbrunner, Kirkpatrick, and Seidel, 1983), β-skeletons (Kirkpatrick and

Radke, 1985), and γ-neighborhood graphs (Veltkamp, 1992) broadened the family by varying scale or the exclusion region. The survey by Jaromczyk and Toussaint (1992) already described applications in morphology, geographic analysis, classification, and computer vision. Today, these mathematical alternatives can be tested under a common software interface.

What Can You Explore?

Recover shapes and holes. A convex hull conceals concavities. Alpha-shapes retain scale-selected Delaunay simplices with straight boundary edges; related alpha-hulls draw circular arcs through the same boundary sites. In our typographic point cloud, decreasing the selected radius changes the two enclosed counters and the connectivity of the letters. Applications range from scanned-object reconstruction to image morphology and exploratory topology; a geometric envelope is not, by itself, a statistical confidence region.

Figure 2 · Image-traced serif Pg · alpha-shape counters

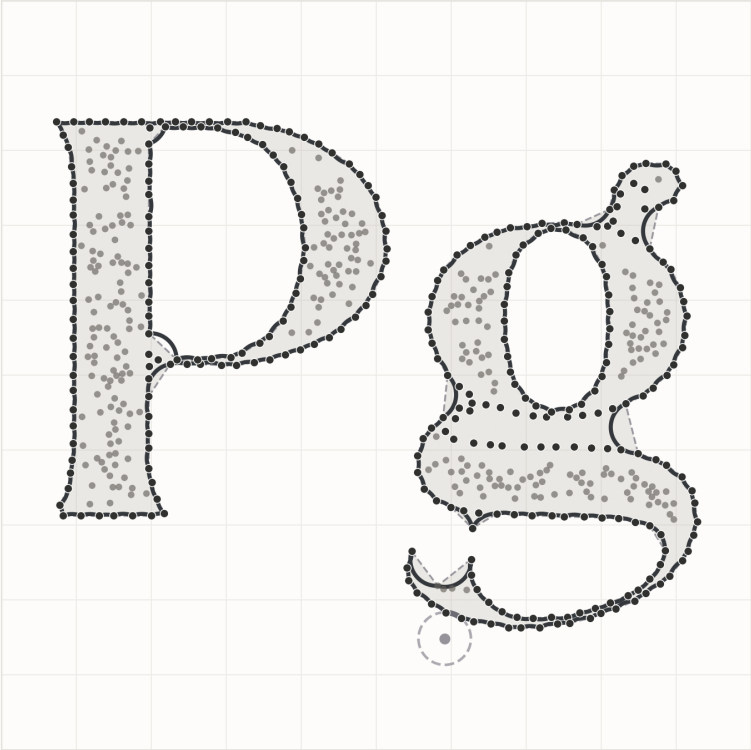


Figure 2: Reconstructing shapes from scattered observations. A synthetic serif *Pg* point cloud illustrates an alpha-shape, circular alpha-hull arcs, and enclosed counters. Change the radius or density, reveal supporting circles, or select a smiley preset. Artwork and samples: the authors. Explore the interactive figure.

Decide whether an edge survives. For an empty-region graph, a pair p, q is adjacent only when a designated region $K(p, q)$ contains no competing point:

$$\{p, q\} \in E_K(P) \quad \Longleftrightarrow \quad K(p, q) \cap (P \setminus \{p, q\}) = \varnothing. \tag{1}$$

Gabriel graphs use diameter disks; RNGs and many β-skeletons use lenses. For generic planar samples, the minimum spanning tree, RNG, Gabriel graph, and Delaunay triangulation obey $\mathrm{MST} \subseteq \mathrm{RNG} \subseteq \mathrm{GG} \subseteq \mathrm{DT}$. Sparser edges can reduce infrastructure length, but not necessarily route detours or communication failures. A drawn sensor link is a geometric candidate, not verified radio coverage.

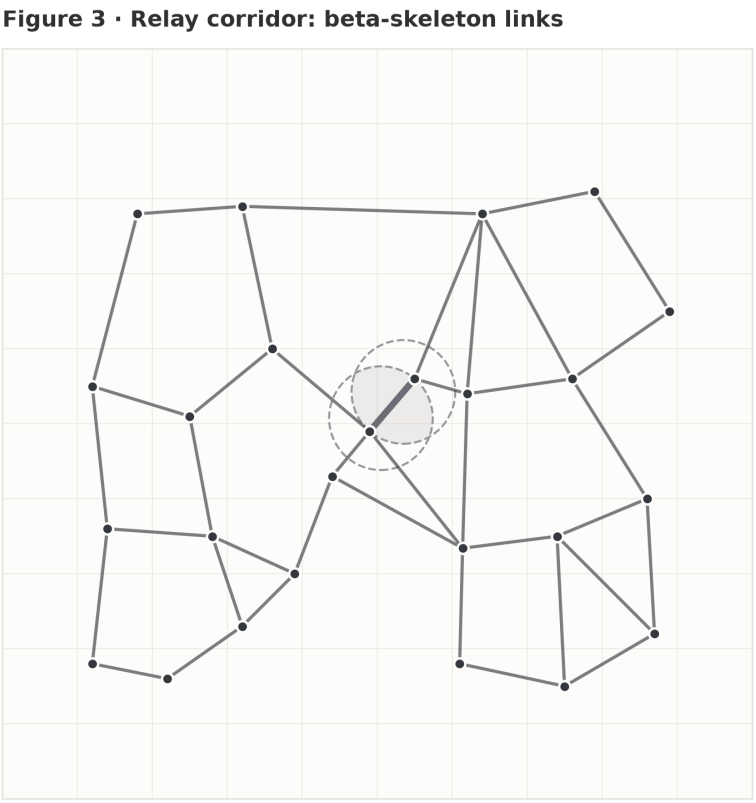


Figure 3: Wireless-network candidate links. Twenty-five synthetic sites in two clusters with relays illustrate a β-skeleton. Select two sites to inspect the defining disks, the common lune, and any point that blocks their connection. Layout and artwork: the authors. Explore the interactive figure.

Compare network designs under controlled conditions. The same point set may produce different total edge lengths, connected components, average degrees, and graph detours. This matters when designing potential road or sensor networks (Aldous and Shun, 2010), reconstructing cyclic street tilings (Maravillo Gómez et al., 2023), or constructing neighborhoods for manifold learning (Dyballa and Zucker, 2023). Figure 4 uses the *same* synthetic locations under Gabriel and RNG rules, then lets visitors replace them with binomial or Poisson samples. Changes reflect the graph rule rather than a different realization.

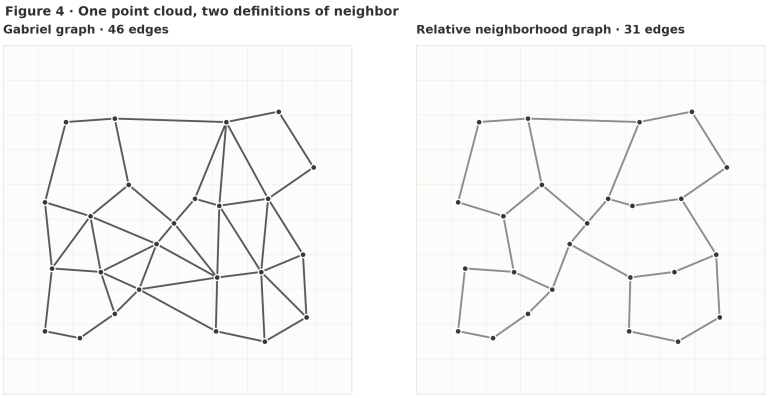


Figure 4: One sample, competing network rules. Gabriel and relative-neighborhood graphs are constructed on identical 25-site synthetic coordinates; compare their edges and densities. Online, select other graph rules or regenerate a common random point sample. Artwork: the authors using ProximityGraphs. Explore the interactive figure.

What Happens When the Points Are Random?

In a stationary Poisson point process of intensity ρ in $\mathbb{R}^d$, suppose a pair at separation r has an exclusion region of volume $a_K r^d$. Its emptiness probability is $\exp(-\rho a_K r^d)$, so the typical mean degree becomes

$$\Pr(\{p, q\} \in E_K \mid r) = e^{-\rho a_K r^d}, \qquad \mathbb{E}[\deg] = \frac{v_d}{a_K}, \tag{2}$$

where v_d is unit-ball volume (Devroye, 1988). Surprisingly, the limiting degree depends on the exclusion-region geometry, not the density. This gives a measurable prediction: the planar Gabriel graph has mean degree 4, while the RNG has approximately 2.56. The theory extends to $\mathbb{R}^d$; the present library and these interactive demonstrations remain primarily planar.

From Theory to an Interactive Python Laboratory

ProximityGraphs provides more than a single graph drawing. Its `SetPoints` tools generate structured and stochastic inputs; constructors cover Voronoi/Delaunay, Gabriel, RNG, nearest-neighbor, minimum-spanning, alpha, β, γ, stepping-stone, and elliptic neighborhood variants. The `Experiment` interface supports seeded repetitions, graph statistics, and comparisons. The current library is distributed on PyPI (`pip install proximitygraphs`), with open source and documented algorithms.

Our **online interactive article** reproduces all four demonstrations as live panels: drag, add, or remove points; reveal geometric construction witnesses; adjust parameters; and export the resulting images. Default scientific scenes are generated by the pinned Python package; edited scenes run an independent browser implementation, explicitly distinguished from the reference. Readers can use the full geometry laboratory to explore further. Proximity graphs do not dictate one correct notion of a neighbor—**they let us make that notion testable.**

References

1. Aldous, D. J., & Shun, J. (2010). Connected spatial networks over random points and a route-length statistic. *Statistical Science, 25*(3), 275–288. doi:10.1214/10-STS335.
2. Devroye, L. (1988). The expected size of some graphs in computational geometry. *Computers & Mathematics with Applications, 15*(1), 53–64.
3. Dyballa, L., & Zucker, S. W. (2023). IAN: Iterated adaptive neighborhoods for manifold learning and dimensionality estimation. *Neural Computation, 35*(3), 453–524. doi:10.1162/neco_a_01566.
4. Edelsbrunner, H., Kirkpatrick, D. G., & Seidel, R. (1983). On the shape of a set of points in the plane. *IEEE Transactions on Information Theory, 29*(4), 551–559. doi:10.1109/TIT.1983.1056714.
5. Gabriel, K. R., & Sokal, R. R. (1969). A new statistical approach to geographic variation analysis. *Systematic Zoology, 18*(3), 259–278. doi:10.2307/2412323.
6. Jaromczyk, J. W., & Toussaint, G. T. (1992). Relative neighborhood graphs and their relatives. *Proceedings of the IEEE, 80*(9), 1502–1517. doi:10.1109/5.163414.
7. Kirkpatrick, D. G., & Radke, J. D. (1985). A framework for computational morphology. In *Computational Geometry* (pp. 217–248). North-Holland. doi:10.1016/B978-0-444-87806-9.50013-X.
8. Maravillo Gómez, H. S., Calvillo Vives, G., & Treviño Aguilar, E. (2023). Recovering cyclic tilings through β-skeletons. *Procedia Computer Science, 223*, 343–351. doi:10.1016/j.procs.2023.08.246.
9. Toussaint, G. T. (1980). The relative neighbourhood graph of a finite planar set. *Pattern Recognition, 12*(4), 261–268.
10. Veltkamp, R. C. (1992). The γ-neighborhood graph. *Computational Geometry: Theory and Applications, 1*(4), 227–246. doi:10.1016/0925-7721(92)90003-B.

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